Major Findings

We separate the major research findings into five areas of *Multimedia Information Processing*, *Networking*, *Databases & Distributed Systems*, *Digital Classroom*, and *Collaborative Computing*. We discuss each area separately.

1 Multimedia Information Processing

1.1 Creating a structured document from handwritten lecture notes

We have developed a system that automatically captures lecture notes in a typical classroom setting, and converts them into a structured document representation that is suitable for annotation and distribution. Textual parts of the notes are separated from non-textual (graphical) parts, and an off-the-shelf handwriting recognition system is used for text recognition. The instructor?s speech is also recorded, and the spoken words are recognized to enhance the text recognition accuracy. A simple-to-use editor is developed to manually edit the structured document to correct any mistakes. The resulting representation is very compact, and requires very low bandwidth for delivery. A Web interface is currently available for browsing lectures on demand.

The E-Beam whiteboard capturing system was purchased for this project. This system consists of special penholders and sensor pods, which can be mounted on any whiteboard, and provides pen color and location information to a host computer. This allows recording and presentation of lectures over a network, as well as online meetings/collaboration.

In addition, we now have a full version of the audio/video logger and face recognition modules from Virage. We have several hours of taped lectures and plan to incorporate these into the above lecture annotation system during the coming year. Data recorded through different devices (such as a camera, audio recorder, white board, and computer based presentations) often are not synchronized. This creates serious problems for event analysis and indexing. We are developing new methods that include speech recognition and handwriting recognition to synchronize the various media.

1.2 Searching for nearest neighbors in high-dimensional feature spaces in relevance feedback

In this work we address the issue of scalability of relevance feedback techniques for large image/video data in high-dimensional feature spaces. Relevance feedback is often used in content based image and video retrieval to support high-level concept queries. During each iteration of a relevance feedback search, the similarity metric is modified based on user's input and a new set of nearest neighbors of a given query vector is computed. This repetitive nearest neighbor computation in high dimensional feature spaces (in hundreds) is expensive, particularly when the number of items in the data set is large (hundreds of thousands). In this context, we present a scheme that exploits correlations between two consecutive nearest neighbor sets and significantly reduces the overall search complexity. Detailed experimental results are provided using a set of about 700,000 image feature vectors. Comparison to the existing method (wherein each iteration is an independent search) shows an order of magnitude overall improvement.

1.3 Lecture on Demand

We have built an infrastructure for the *lecture on demand* application. This infrastructure consists of a video capture card, the Virage VideoLogger system, our Java-based indexing framework called jetMAX, and the Virage Video Application Server. For a recorded lecture, we first digitize the video signal using the capture card and transform it into an MPEG-2 stream. The MPEG-2 files are then read and analyzed by the VideoLogger system. In addition to the included audio analysis, we implemented two new analysis plug-ins for color and texture feature extraction. The extraction of all features happens in real-time. Therefore, the analysis of a lecture video can be completed within 1-2 hours and results in about 5MB of raw vector and text data. This data is then stored in our indexing framework jetMAX which supports a variety of indexing schemes, such as R-trees and X-trees. In order to process user queries, we facilitate Virage's Video Application Server that provides web-based multi media search and retrieval interfaces. For a given user query, we issue the corresponding subqueries to the index structures for color, texture, etc. and combine th e results. The answer is then presented to the user as a list of video shots ranked by relevance from which she can pick the ones to be played back. The playback of the videos is also handled by the Video Application Server. Note that all videos are stored on a separate video server machine independent of the indexing system. Future work includes the use of prediction models to support faster queries and adaptive page layouts on disks.

1.4 New perceptual distance function

The team has discovered a perceptual distance function that works substantially better than the traditional Minkowski-like functions. This function was formulated after conducting extensive data mining and has shown strong link to similarity theories in cognitive science. We believe this result will have profound impact on the CBIR community.

1.5 Virtual I/O

The team has invented Virtual I/O, which drastically improves the preemptibility of an I/O request. Consequently, a high priority request that requires short response time can be serviced by the disk promptly, usually within one millisecond. This work will enable an array of disk-based interactive multimedia applications.

2 Networking

Our work in networking has following two general directions: (i) hybrid networks, that integrate both infrastructure and ad hoc network functionality, and (ii) quality of service for wireless data.

2.1 Hybrid networks

We have made significant strides in creating hybrid wireless networks. These networks integrate infrastructured wireless and ad hoc wireless network technology. An infrastructured wireless network is a wireless network with access points distributed along a wired backbone, and mobile nodes that communicate directly with the access points. An ad hoc network is a network comprised solely of mobile wireless nodes. There is no wired backbone; nodes communicate directly with one another and often serve as relays for data packet forwarding. By integrating these two technologies, multi-hop paths between mobile nodes can be discovered to access points so that Internet connectivity can be attained. Hybrid networks ease deployment of infrastructured wireless networks by relaxing the access point placement constraints. Because

mobile nodes no longer have to be within direct transmission range of access points to attain Internet connectivity, access points can be more freely placed and the worry of dead zones is eliminated.

Our work in hybrid networks has investigated networking and address autoconfiguration issues. The networking research entailed developing a routing protocol to discover routes from nodes within the ad hoc network to nodes within the wired Internet. For this solution we integrated the Mobile IP and Ad hoc On-Demand Distance Vector (AODV) routing protocol. Mobile IP enables roaming away from a home network, while AODV allows for the discovery of routes within ad hoc networks. Experiments with the system show that our solution provides for high data throughput without the introduction of significant overhead. Address autoconfiguration is another important component of such a hybrid system. Address autoconfiguration allows mobile nodes to obtain unique IP addresses in ad hoc networks where a DHCP server does not exist, or to obtain co-located care-of address in Mobile IP networks where a foreign agent is not available. We have developed a solution whereby mobile nodes can obtain unique IP addresses in each of these scenarios, and can continue to guarantee the address uniqueness even after network partitions and merges.

2.2 QoS for wireless data

An important new area of investigation is that of defining a notion of Quality of Service (QoS) for wireless data, and to obtain scheduling strategies for optimizing QoS, accounting for heavy-tailed characteristics of Internet traffic. Instead of the conventional wisdom of provisioning more conservatively for heavy-tailed traffic, we employ scheduling strategies that implicitly penalize the long transactions (e.g., long file transfers) that contribute to the heavy tails, while providing good performance to short transactions (e.g., a short web page download), without actually requiring knowledge of the transaction length. We have developed a transaction-based QoS framework, in which an exemplary service guarantee might be: 99the transactions will see a rate of 50 Kbps. For a typical heavy-tailed distribution, giving the preceding guarantee while maintaining 90 if the conventional Round Robin (RR) scheduler is used, and a capacity of only 250 Kbps if the proposed scheduling strategy is used. Extensions of these ideas to a wireless LAN setting, and interactions of the scheduling methods with end-to-end transport mechanisms, are currently being investigated.

In addition to this work, a significant technology transfer effort, working with Bytemobile, Inc., a wireless networking start-up, was undertaken. Methods for optimizing performance for wireless data users, loosely inspired by ideas from our research on end-to-end transport, were incorporated into Bytemobile's network infrastructure node, which is now deployed at major wireless carriers such as Sprint and Vodafone. In physical layer wireless research, new methods for equalization and multiuser detection were developed. Current physical layer research efforts, which include close collaboration with Motorola, focus on understanding and attaining the fundamental limits of wideband wireless channels, taking into account the variations in time and frequency inherent to such channels. Technology transfer efforts will include application of the ideas developed into Motorola Labs' experimental high-speed, outdoor, wireless Orthogonal Frequency Division Multiplexing (OFDM) system.

3 Databases & Distributed Systems

3.1 Support for cluster-based online services

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The Neptune project is studying data replication, load-balancing, fault tolerance, and QoS support for cluster-based online services. It provides a flexible interface to glue and replicate existing service modules

and accommodates a variety of underlying storage mechanisms. Neptune maintains dynamic and location-transparent mapping to isolate faulty service modules and enforce replica consistency. It allows efficient use of multiple-level replication consistency mechanisms with staleness control and failure recovery support. Neptune has been successfully deployed in www.teoma.com and www.ask.com for managing large service clusters.

Load balancing on a cluster of machines has been studied extensively in the literature, mainly focusing on coarse-grain distributed computation. Fine-grain services introduce additional challenges because system states fluctuate rapidly for those services and system performance is highly sensitive to various overhead. Through simulations and a prototype system implementation in a Linux cluster, our study concludes that: 1) Random polling based load-balancing policies are well-suited for fine-grain network services; 2) A small poll size provides sufficient information for load balancing, while an excessively large poll size may in fact degrade the performance due to polling overhead; 3) Discarding slow-responding polls can further improve system performance. Quality of service (QoS) support that provides customized service qualities to multiple classes of client requests can effectively utilize available system resources. Our QoS framework strives to achieve four goals. First, the framework provides a flexible mechanism for service providers to specify a variety of desired service quality metrics through QoS yield functions. Secondly, the framework provisions service differentiation and admission control for multiple classes of service accesses by employing both the yield functions and resource allocation guarantees. Thirdly, the framework achieves efficient resource utilization by producing high aggregate QoS yield through greedy scheduling and speculative admission control that drops zero or low yield requests. Finally, the design of this QoS framework takes into consideration service scalability and availability by deploying a decentralized two-level request distribution and scheduling scheme.

3.2 Mobile middleware

Mobility opens up new paradigms for interaction in large systems. Specifically, there is a need for dynamic peer-to-peer services in which agile devices can intermittently connect, compute, and disconnect with each other. In order to support such interactions, we need a middleware which can place computations and data adaptively based on the network condition and configuration, and which can store persistent objects and computations. Furthermore, the system should support weak levels of consistency so that a mobile user can adapt and choose the appropriate consistency guarantees from shared objects and other users. Finally, the system should support a powerful way to merge disconnected computations so that divergent histories can be merged in a consistent manner.

We have developed a prefetching technique that is able to accurately prefetch complex pointer-based data structures. Compile time shape analysis is used to determine the set of possible future objects accessed by a program. We use the reachability graph generated during the shape analysis to create prefetching schedules for selected program points. The program is annotated to visit the reachability graph with the set of accessible objects at any program point, generating a superset of future object references. The technique is applicable to both static and dynamically created object structures. We demonstrate this technique in the context of mobile computing and show a speedup for a variety of Java benchmarks in the presence of disconnections.

3.3 New Index Structures

Rapid advances in information technology have led to increased demand for the storage, management, access, and use of multimedia objects. We are designing new index structures, novel querying algorithms, and constructing a distributed indexing infrastructure. We have developed a new technique for translating costly high-dimensional NN and All-NN-queries into efficiently executable range queries. Our approach is

based on the fractal dimensionality of the dataset and a data sample for predicting the NN-radius. The technique is very general and can be applied to any page-based index structure. Our experimental results indicate a speedup of up to 5 for NN-queries and up to 4 for All-NN-queries compared to traditional approaches. We have developed a new index structure that allows similarity search for multi-attribute time-series data. This technique can handle shift and scale invariance. The index structure recursively splits the data space based on the angular and Euclidean distance. Initial performance results indicate a speedup of 5 to 45 times over competing techniques. We have also developed new index structures for DNA strings and interactive query algorithms that provide a fast response time.

4 Digital Classroom

4.1 Enhancements to Classroom Capabilities

Significant changes have been made to enhance audio and video capabilities in the classroom.

Audio Upgrades. Initially audio capture was limited to that of the presenter using a wireless lapel microphone, and a small mixer. Another challenging problem was that the audio received from remote sites was often picked up by the presenter's lapel mic, re-encoded, and sent back to the remote site, creating echo. This also presented a challenge to our efforts add microphones for the capture of audience audio. In order to solve these problems we installed a 12 channel digital mixer with distributed echo cancellation. This was the most dynamic solution available. Because echo cancellation is distributed for each of the microphone inputs, it can be fine tuned for each microphone. The mixer provides a high level of control including automatic gain control, the ability to gate mics on and off based on volume levels, and the option to add filters and otherwise fine tune the setting of each audio channel. The mixer enables the routing of audio for different input/output configurations. For example it is possible to send one channel which includes all audio coming from inside the room to be encoded for remote sites, while another channel containing all of the room's audio as well as that of remote sites from the access grid can be sent to an encoding machine for streaming and/or archiving. The audio can be encoded in a variety of different formats using RAT (Robust Audio Tool) including high quality linear PCM, and can also be sent along with video for encoding in one of the major proprietary formats.

Video Upgrades. Our initial infrastructure included a high quality (16 input, 12 output) video switch. All video and computer feeds including three cameras, presenter laptop, VCR, and operator display machines (used for viewing remote participants), are sent to the video switch inputs. This provides control over where those streams are routed. Outputs include the room's three data projectors, and PCs for encoding and streaming in a variety of formats. Using audio/video signal splitter amplifiers we are able to take each camera's video and send copies to multiple encoding machines (along with audio). By linking together all three cameras in the room, centralizing camera control, and adding a second small video switch, it is now possible to edit video in real time. Using the cameras tilt/zoom, and preset functions it is possible to follow speaker's movements, zoom in on audience members, and make smooth cuts between the different cameras. This allows for the creation of high quality edited video for the both real time streaming and archiving, with video encoded into one or more different formats including all of the major proprietary formats (REAL, Quicktime, Media Player, NetMeeting). At the same time the system can transmit all three video streams using VIC (which encodes in MPEG, Motion JPEG, H.323, and additional formats) to remote sites via multicast, providing them with three different video streams of the classroom, which can include the presenter, audience or even the room's operators.

4.2 Classroom as an Access Grid Node

The upgrades made to audio and video equipment fit specifications for the Access Grid (AG), a community of similarly equipped "nodes" utilizing Access Grid virtual venues software to do high quality audio/video conferencing over Internet2 using multicast. This has been particularly useful in testing the room because of the large, active, participant base in the AG community. The AG model also offers valuable additional conferencing capabilities including distributed Powerpoint (DPPT), and a back-channel method of text communication for "node" operators to stay in contact without disrupting a meeting or presentation. The AG technology, combined with the room's existing technology, has helped to make the classroom an extremely versatile and effective hub for both production of audio/video content as well as real time collaborative learning and research experiments.

4.3 Created Secondary Classroom

We have also worked to complete a second classroom. Not only does this give us an additional site to coordinate with, it also provides a consistent, reliable way to test various classroom activities locally. We have set up smaller secondary classroom in the Computer Science conference room. This project also allowed for the realization of a compact or "mini-node" which is much less resource intensive. Utilizing the same technology as in the classroom with scaled down audio and video capability we have been able to create a satellite location for participating in classroom presentations, and Access Grid meetings. This room, although less dynamic then the primary classroom, is well equipped as a satellite location for audience participation but is also quite capable of being the site of a presentation. Although it has only one camera and fewer microphones, it has the same capability to encode in multiple formats, run distributed Powerpoint, and send audio and video of both the presenter and the audience. This room is therefore a fully functional production environment, but also offers valuable access to instant testing and troubleshooting of the primary classroom.

4.4 Recording Classroom Activities

One of the most significant uses of the Digital Classroom is to archive the content presented in the digital class room. We used the room for the delivery of instructional courses and recorded the lectures in multiple formats in real-time for presentation to the multimedia processing group. Because of the versatility in the prerequisites of various multimedia information processing techniques the lectures had to be recorded in various formats like Real media format, DV, MPEG-1 and MPEG-2. Incompatibility issues with hardware made it necessary to record the media in these formats in parallel, which we dealt with by trying out commercially available solutions.

The goal was to produce media stream in a resolution high enough to be used as the source material for extensive multimedia filtering and editing. The speech recognition and video indexing process requires the input media in the MPEG-2 format while the media format which is suitable for viewing on client desktops has to be one of the major proprietary media formats (such as REAL, Quicktime, or Media Player). So archiving was done keeping in mind various future needs.

4.5 Building a Classroom Services Architecture

Collaborative Virtual Environments (CVEs) enable multiple users in different physical locations to share a single "virtual space" to communicate and collaborate as if they actually shared the same physical space. Such environments provide rich, interactive audio-visual content to foster a feeling of presence between remote participants. Because of their convenience and cost-effectiveness, CVEs are emerging as a

dominant communication paradigm that finds applicability in numerous domains of strategic importance to Motorola such as public safety.

Such virtual environments have three core limitations. First, they have undifferentiated content. All participants receive all content; this often causes device or sensory information overload. Second, they are not personalizable. Individual participants cannot enhance their experience despite possessing additional resources. Finally, they are not mobile-friendly. "Untethered" devices (such as cellphones and handhelds) are difficult to integrate into broadband-enabled CVEs. Not only is mobility not supported conceptually, but the often numerous audio and video feeds overwhelm the devices' limited resources.

Because CVEs are becoming ubiquitous, substantial value can be derived by addressing the needs above. A service-based architecture can alleviate these problems. It can also do so in a modular manner applicable to a single physical space and extensible to a shared virtual space. Our work is beginning to target the design of a service architecture, building it for a single physical space, and implementing core services to integrate wireless mobile devices into the space. The architecture is being deployed and evaluated in the digital classroom. The hope is then to extend it to public safety or other applications.

4.6 Content Distribution in the Classroom

Using the World Wide Web as a core component of the digital classroom serves to facilitate interaction and content distribution. Moreover, the traditional pull-based method of content delivery is insufficient for applications in the digital classroom. In our work, we have focused on developing a model to implement scalable, push-based information dissemination applications. Our goal is to move away from traditional web techniques because they have a number of limitations. First, the repeated one-to-one (unicast) delivery of large information bases is wasteful of server and network resources. Second, the current web model supports only limited user-to-user interaction. Our approach has attempted to overcome these limitations. At the heart of our model is an interactive, dynamic information base that allows users to interact not only with the system, but with each other. In addition, our model provides multicast-based support for large-scale data delivery to a large user base. In this way, the system can handle large numbers of requests without becoming overloaded.

Our model partially leverages the properties of the *Interactive Multimedia Jukebox* (IMJ). The IMJ is a novel architecture that addresses the issues of scalability and flexibility. The IMJ delivers on-demand programs via multicast and uses a WWW-based interface to schedule requests using a jukebox paradigm. Our goal is to generalize the IMJ model to develop a new model that achieves the IMJ's benefits of scalability and flexibility while supporting a variety of systems beyond streamed video-on-demand. Our approach, called the *Active Information System* (AIS), has proven useful in developing content storage and retrieval in the digital classroom. In particular, we can support the archival, storage, and access of classes. Furthermore, we can create distributed office hours by using the AIS to schedule and support distributed conferencing between instructors, teaching assistants, and students.

5 Collaborative Computing

5.1 Molecular visualization tool

We have developed a molecular visualization tool, which helps remote researchers collaborate by synchronously examining the same protein molecule. The collaboration is facilitated by sharing the visual representations of the molecule and by providing a multimedia annotation mechanism for sharing the ideas. Our implementation is based on Java 3D, thus providing compatibility among different systems and having the ability to be run remotely through web browsers. However, using Java 3D for visualization may bring out some problems with it. The primary concern about molecular visualization tools based on

Java 3D are in their being slow in terms of interaction speed and in their inability to load large molecules. We present techniques to tackle these problems and demonstrate that our optimization techniques achieve an interactive rendering speed twice faster and are able to load molecules three times larger than other systems based on Java 3D.

5.2 Sticky notes

We developed an annotation tool, "sticky notes," for facilitating distributed collaboration for 3D modeling and visualization. Sticky notes are small 3D objects that coexist with and augment the host 3D objects. Sticky notes provide a convenient way for people to annotate (with both text and audio) a model. However, our note mechanisms go beyond simple passive annotation to also provide active interactivity. We introduce the action notes that perform operations on-demand, and the active notes that monitor the environment for specific events and automatically handle such events when they occur without the user intervention. Sticky notes also exhibit levels-of-detail behaviors that adapt their appearance for optimal viewing. As a demonstration of concept, we use sticky notes for an application in collaborative visualization of protein data from the Protein Data Bank. The application environment centers on a "virtual lab," which comprises a number of "virtual workbenches." Each workbench accommodates a group of geographically distributed users collaborating on a protein model. The users of a workbench can annotate the model with sticky notes and share these notes. Sticky notes thus allow sharing of ideas, enable joint attention, and facilitate group work.